Cost Analysis

2026-04-06

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1 Assumptions

The aim of the analysis is to identify the most cost-effective production periods in months of data and recommend optimal operating windows for the corresponding key influencers that can be set as a benchmark for the weekly analysis.

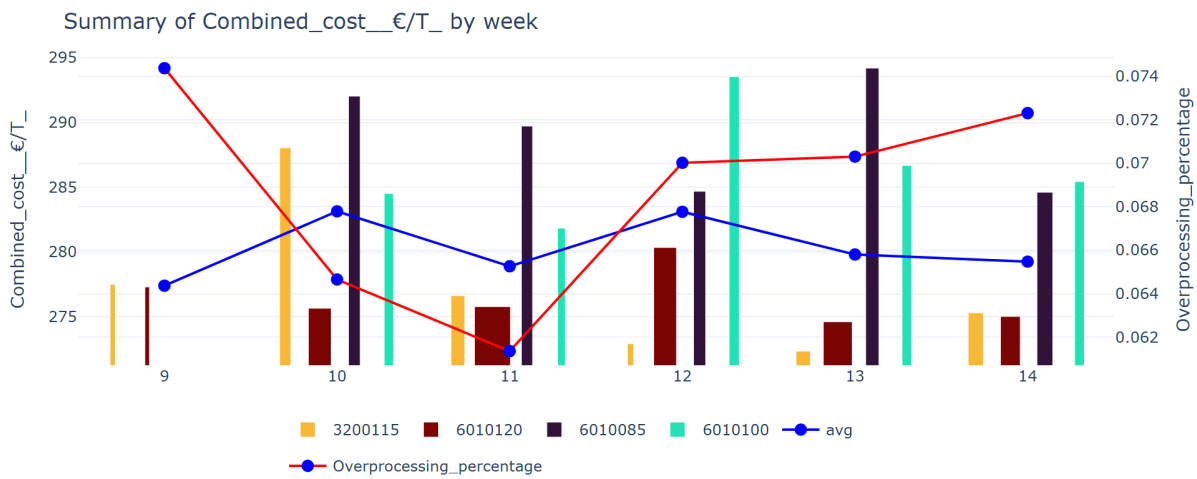
- **Period Analyzed**: March 01, 2026 – April 06, 2026
- **Cost Components Included**: Fibre, steam, electricity, starch, and chemicals.
- **Grades Included**: Only grades 6010085, 6010100, 601020 and 3200115 are included in this week's analysis.
- **Data Selection Criteria**: Only operating conditions where strength properties exceed specification thresholds were considered.
- **Naming Conventions Used**:
  - **Current**: Refers to the last week of the analyzed period.

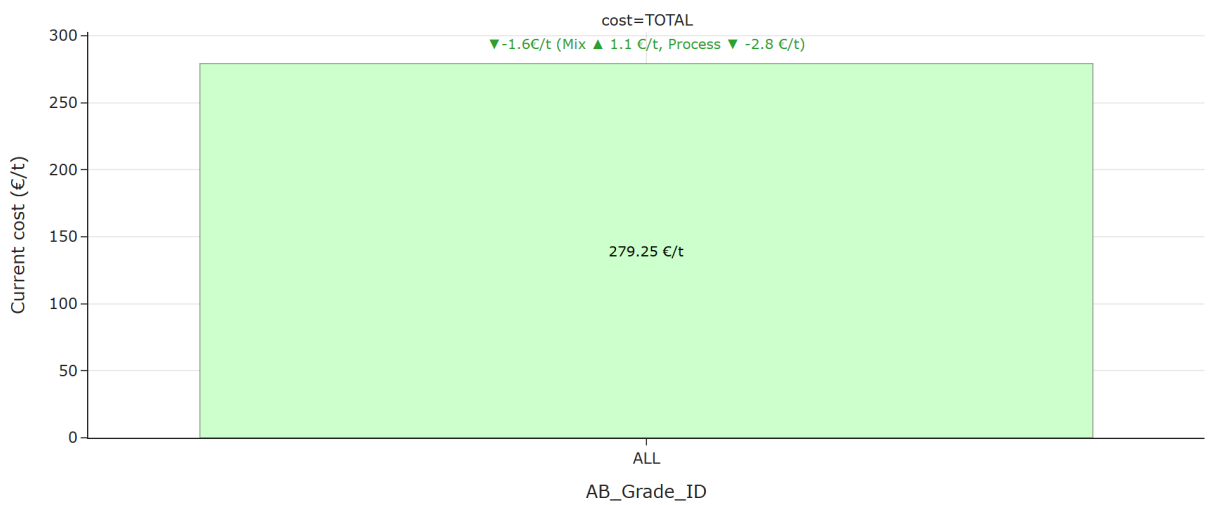
- **Historic**: Represents the average cost over the monthly interval previous to the "Current" period. Used as baseline.
- **Best**: Corresponds to the cost associated with the optimal batch configuration.

2 Executive Summary

The following plot illustrates how total cost varies across different grades, enabling a comparison between the current configuration and the historical baseline.

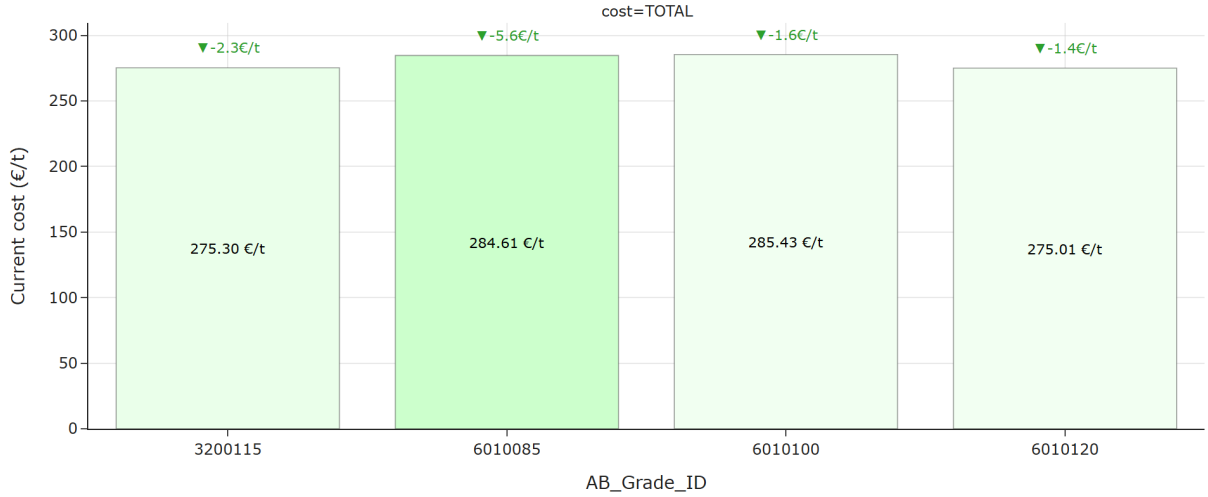
The next plot displays how total cost evolves over time, with a breakdown by grade to show individual contributions.





For all considered grades, TOTAL cost saw a decrease of 1.62 €/t, resulting in 279.25 €/t.

Breaking down the overall cost change, +1.14 €/t was driven by a more expensive mix of grades, and -2.76 €/t was due to an improvement in the process.



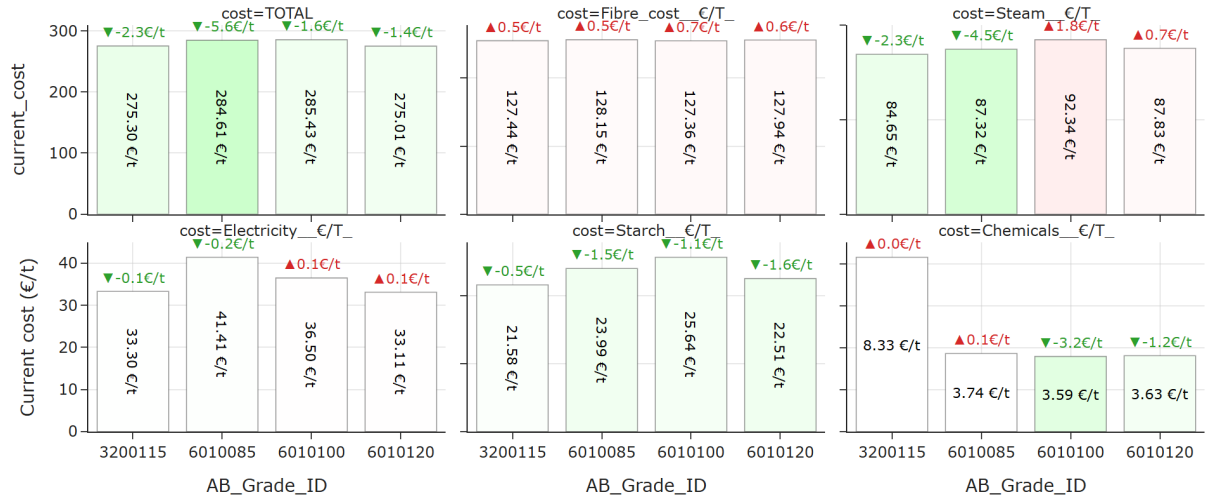
For grade 3200115, TOTAL cost saw a decrease of 2.29 €/t, resulting in 275.30 €/t.

For grade 6010085, TOTAL cost saw a decrease of 5.62 €/t, resulting in 284.61 €/t.

For grade 6010100, TOTAL cost saw a decrease of 1.60 €/t, resulting in 285.43 €/t.

For grade 6010120, TOTAL cost saw a decrease of 1.41 €/t, resulting in 275.01 €/t.

Looking across individual grades, the strongest cost decrease was observed for grade 6010085 (-5.62 €/t). These movements stand out and may require further investigation.

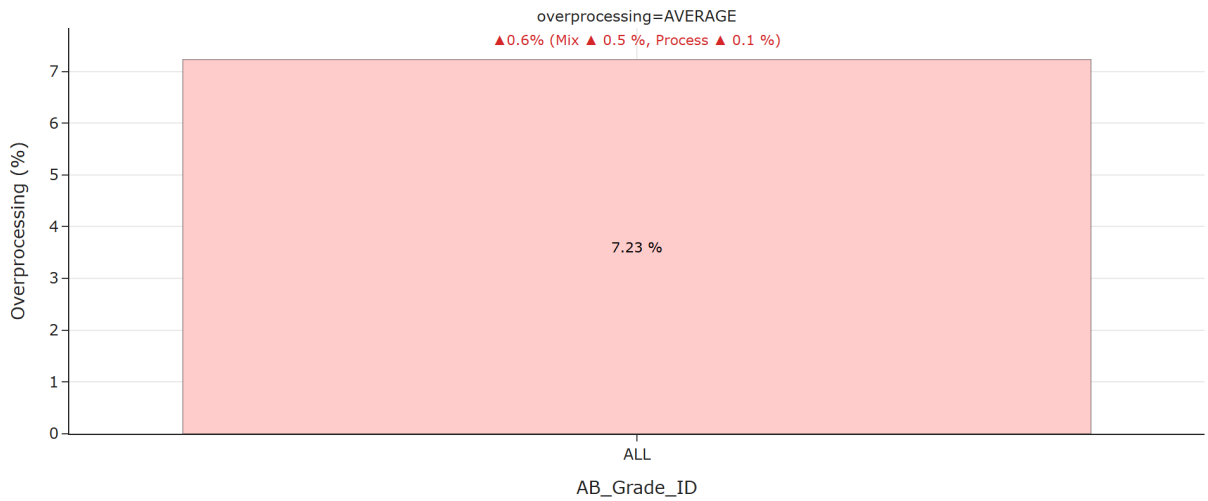


Average change by component:

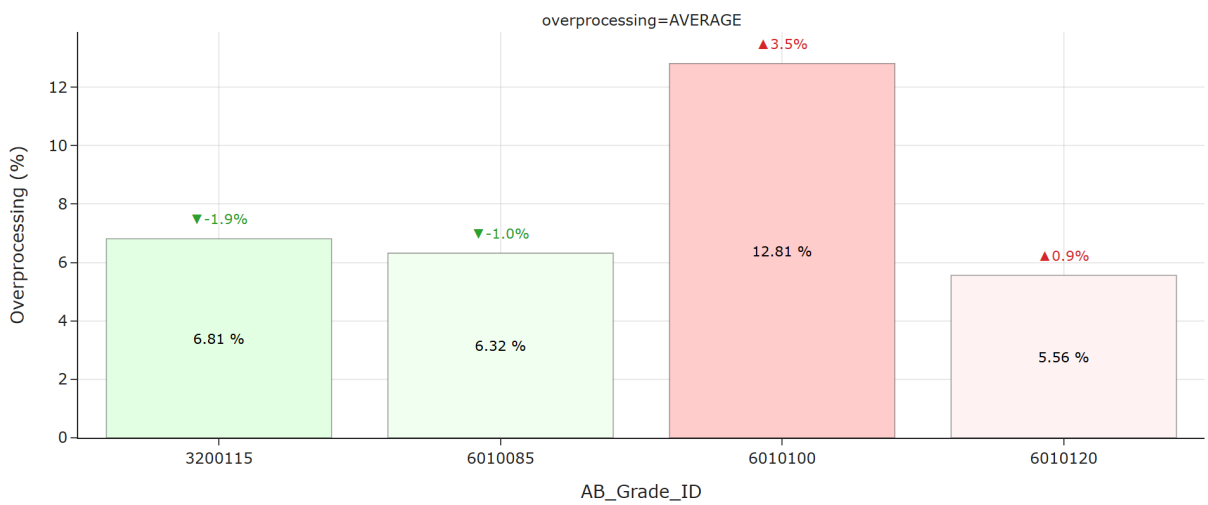
- **Starch_€/T** decreased on average (-1.17 €/t).
- **Chemicals_€/T** decreased on average (-1.07 €/t).

- **Steam_€/T** decreased on average (-1.07 €/t).
- **Fibre_cost_€/T** increased on average (+0.60 €/t).
- **Electricity_€/T** decreased on average (-0.01 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Fibre_cost_€/T** (+0.60 €/t on average), with the highest increase in grade 6010100 (+0.70 €/t) and the largest average decrease comes from **Starch_€/T** (-1.17 €/t on average), with the biggest drop in grade 6010120 (-1.61 €/t). These components stand out and may require further investigation.



For all considered grades, TOTAL overprocessing saw an increase of 0.56 %, resulting in 7.23 %.



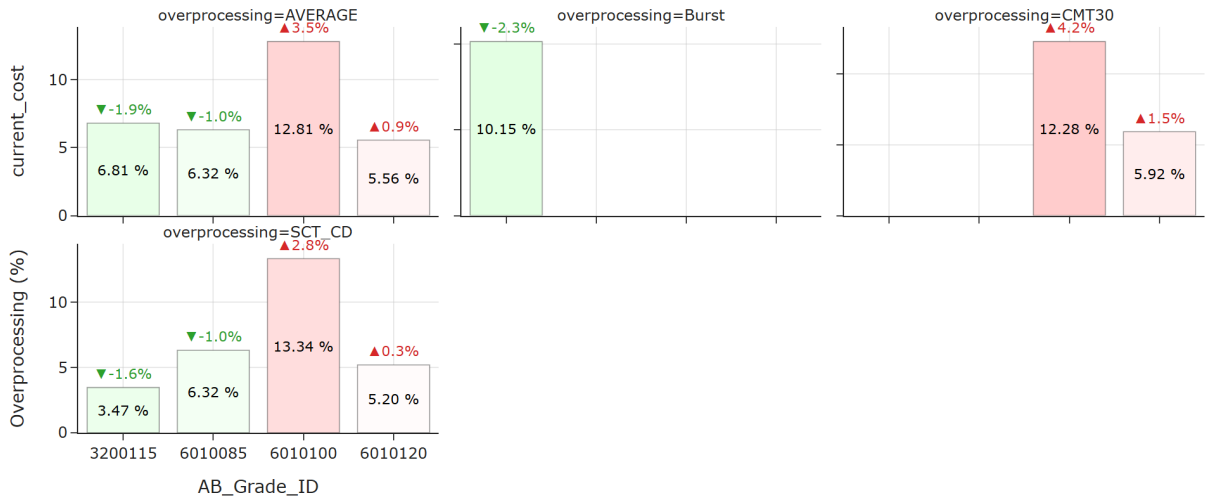
For grade 3200115, TOTAL overprocessing saw a decrease of 1.95 %, resulting in 6.81 %.

For grade 6010085, TOTAL overprocessing saw a decrease of 0.99 %, resulting in 6.32 %.

For grade 6010100, TOTAL overprocessing saw an increase of 3.52 %, resulting in 12.81 %.

For grade 6010120, TOTAL overprocessing saw an increase of 0.90 %, resulting in 5.56 %.

Looking across individual grades, the strongest overprocessing increase was observed for grade 6010100 (+3.52 %) and the strongest overprocessing decrease was observed for grade 3200115 (-1.95 %). These movements stand out and may require further investigation.



Average change by component:

- **CMT30** increased on average (+2.84 %).
- **Burst** decreased on average (-2.33 %).
- **SCT_CD** increased on average (+0.15 %).

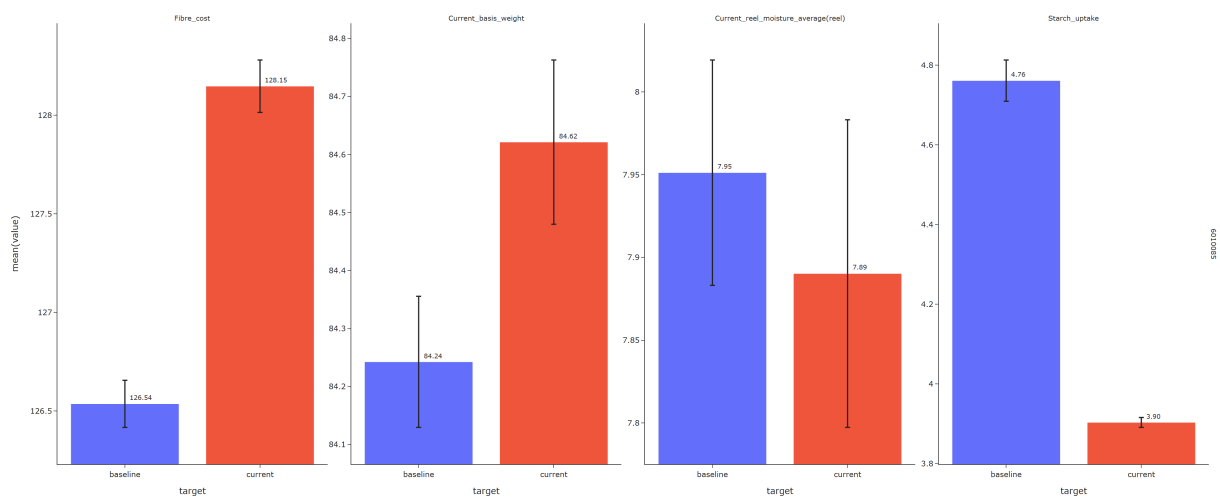
A more detailed breakdown by component shows that the largest average increase comes from **CMT30** (+2.84 % on average), with the highest increase in grade 6010100 (+4.23 %) and the largest average decrease comes from **Burst** (-2.33 % on average), with the biggest drop in grade 3200115 (-2.33 %). These components stand out and may require further investigation.

3 Breakdown by grades

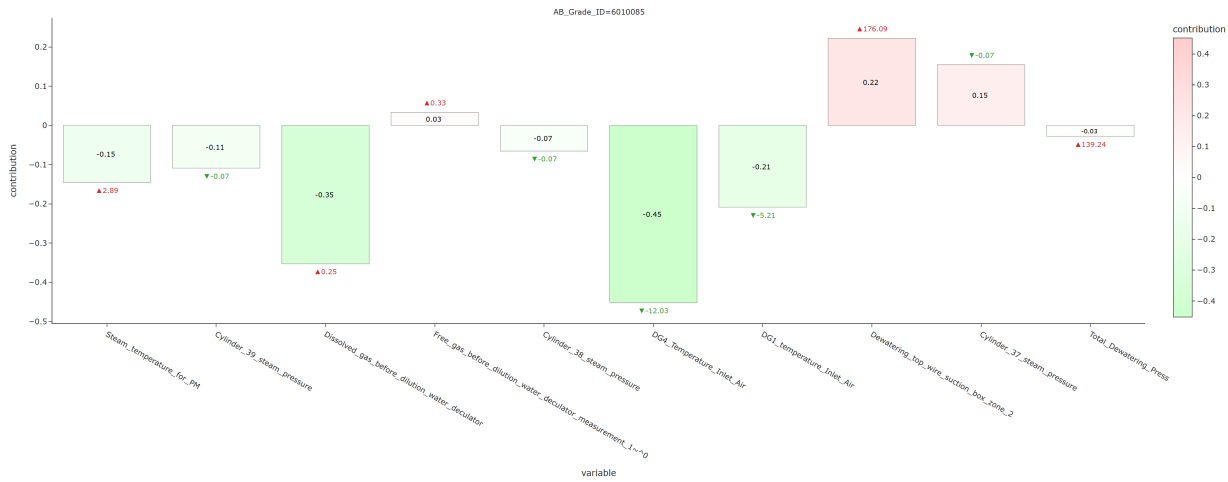
3.1 6010085

3.1.1 Fiber cost

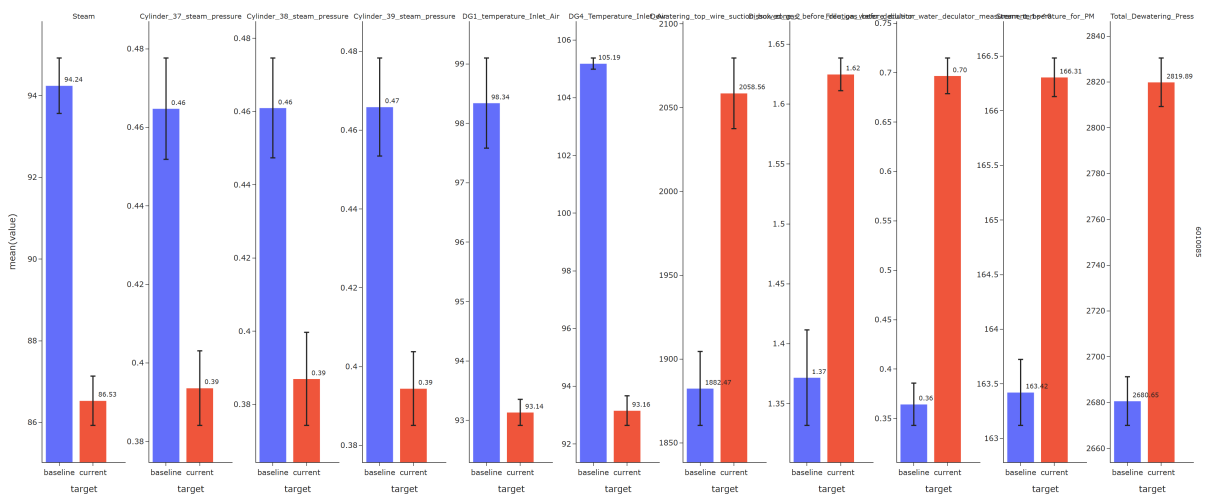
For grade 6010085, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-increasing drivers: - Starch uptake g/m2 decreased (-0.86)



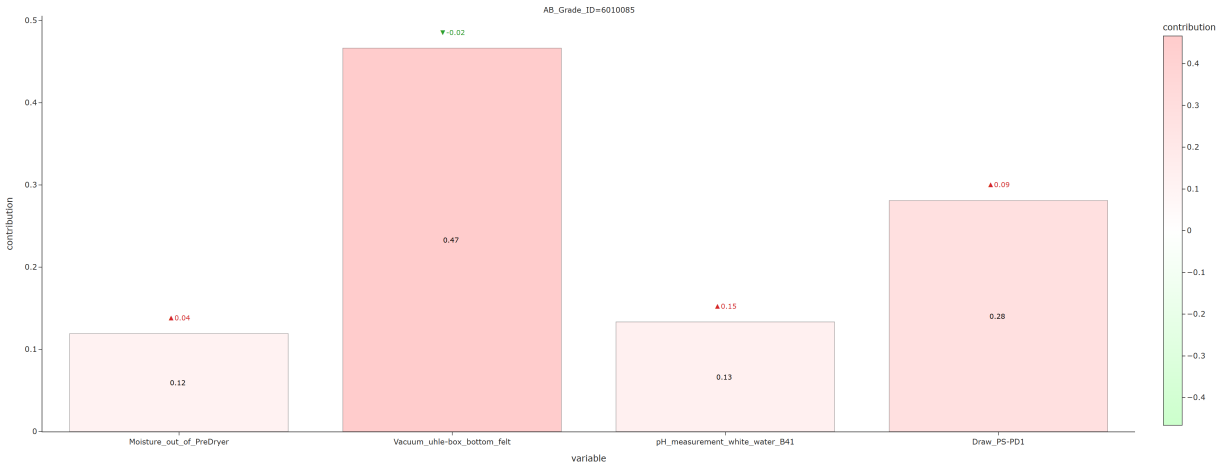
3.1.2 Steam cost



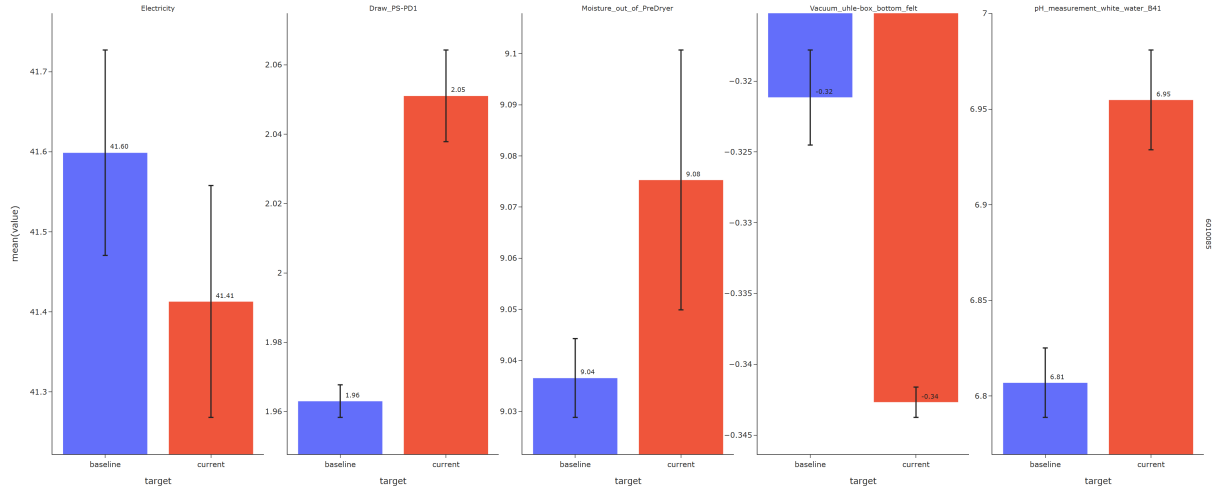
For grade 6010085, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - DG4 Temperature Inlet Air decreased (-12.03) - Dissolved gas before dilution water deculator increased (+0.25) Cost-increasing drivers: - Dewatering top wire suction box zone 2 increased (+176.09)



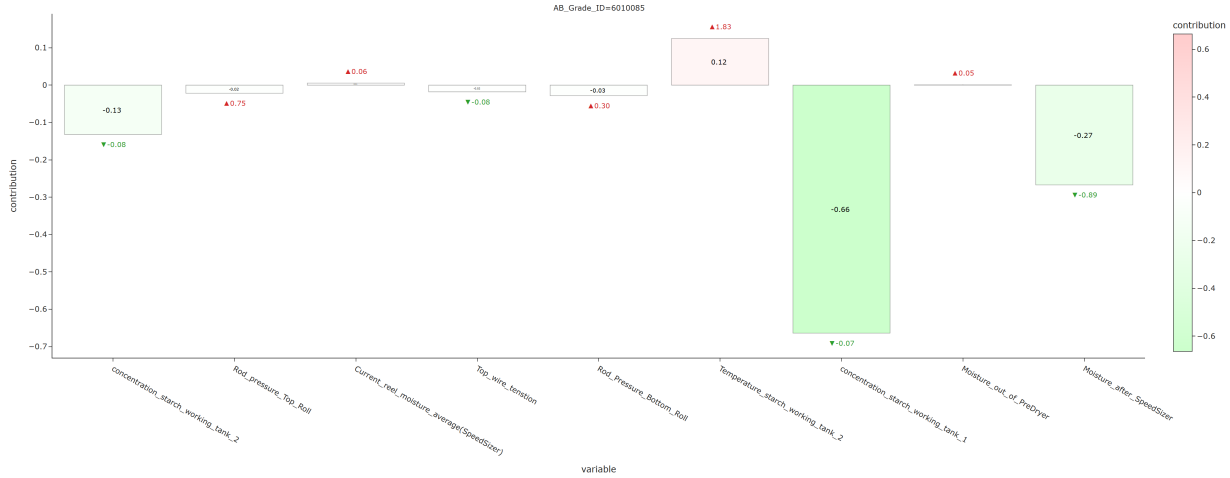
3.1.3 Electricity cost



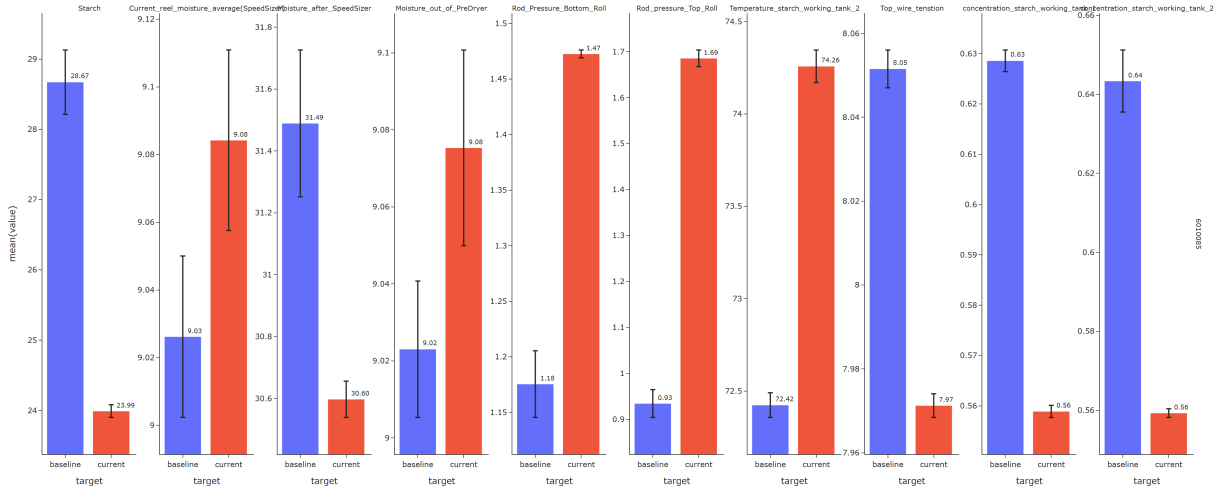
For grade 6010085, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-increasing drivers: - Vacuum uhle-box bottom felt decreased (-0.02)



3.1.4 Starch cost



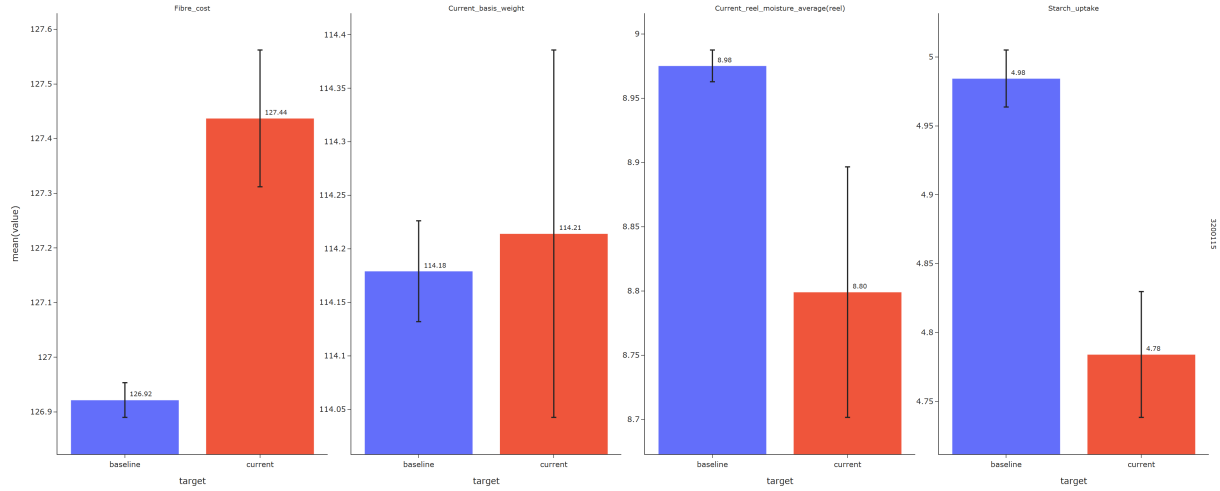
For grade 6010085, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Concentration starch working tank 1 decreased (-0.07) - Moisture after SpeedSizer decreased (-0.89)



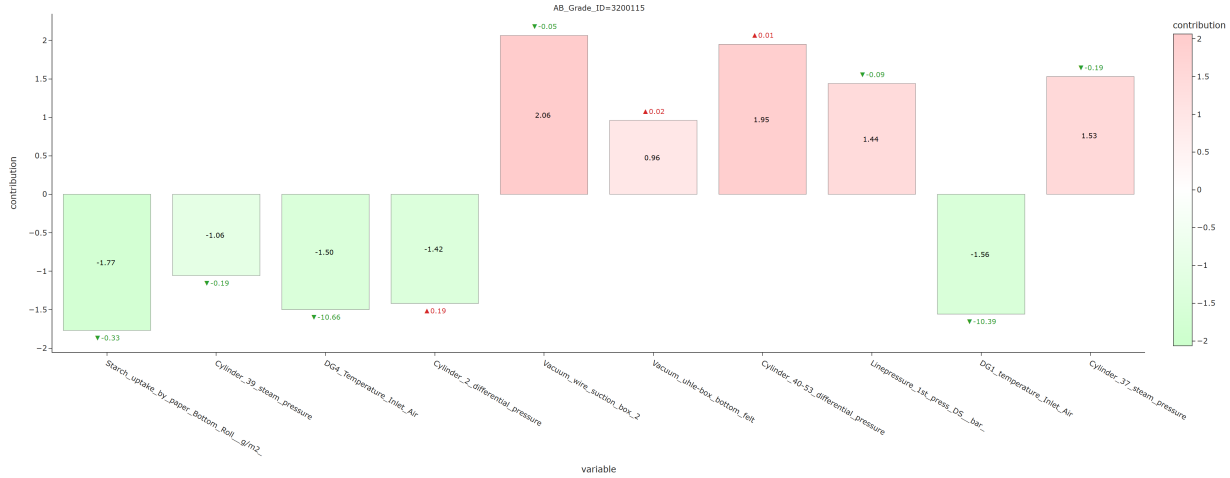
3.2 3200115

3.2.1 Fiber cost

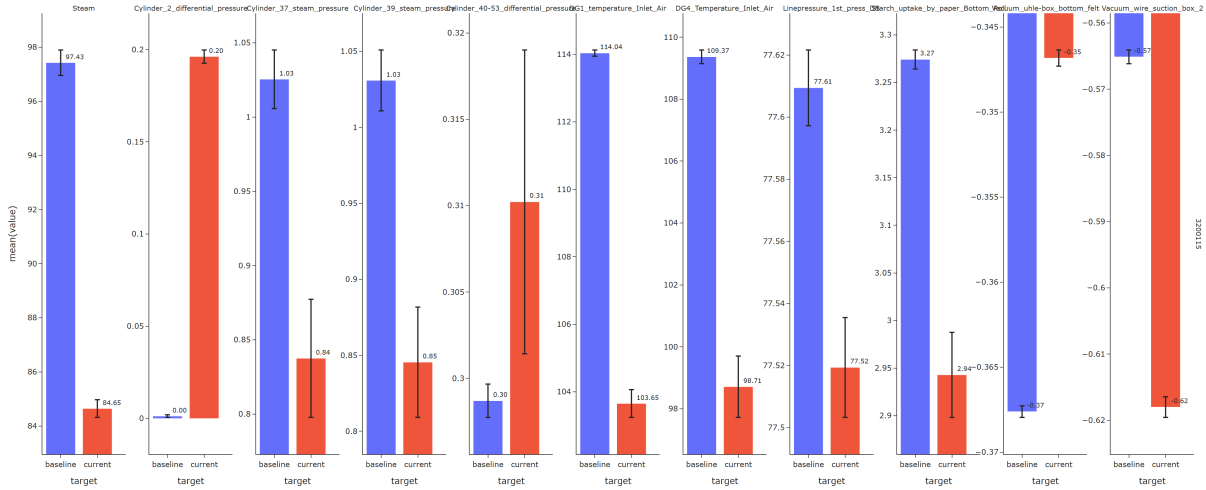
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-increasing drivers: - Current reel moisture average(reel) decreased (-0.18)



3.2.2 Steam cost

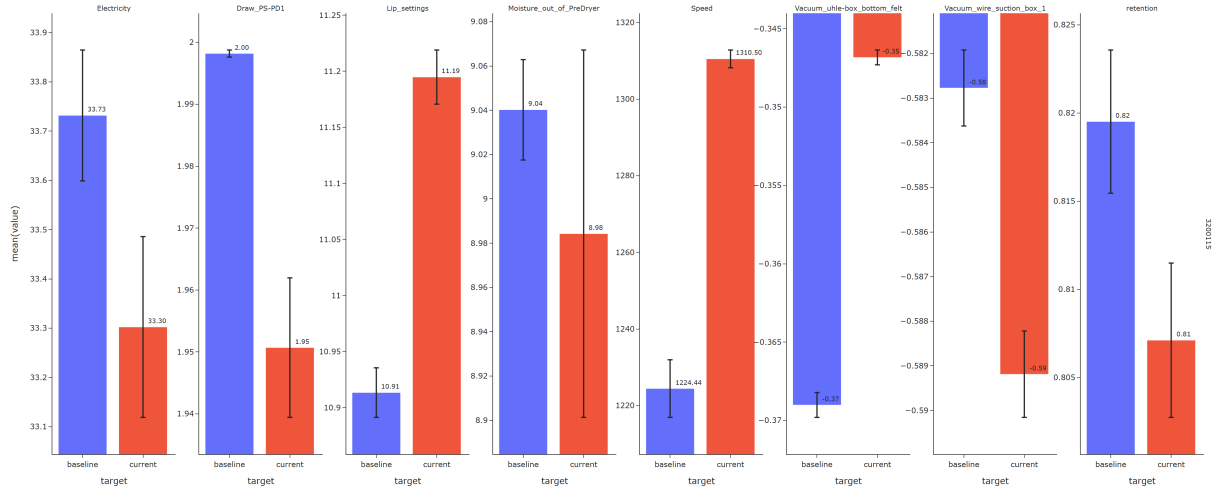


For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Starch uptake by paper Bottom Roll g/m2 decreased (-0.33) - DG1 temperature Inlet Air decreased (-10.39) Cost-increasing drivers: - Vacuum wire suction box 2 decreased (-0.05) - Cylinder 40-53 differential pressure increased (+0.01)

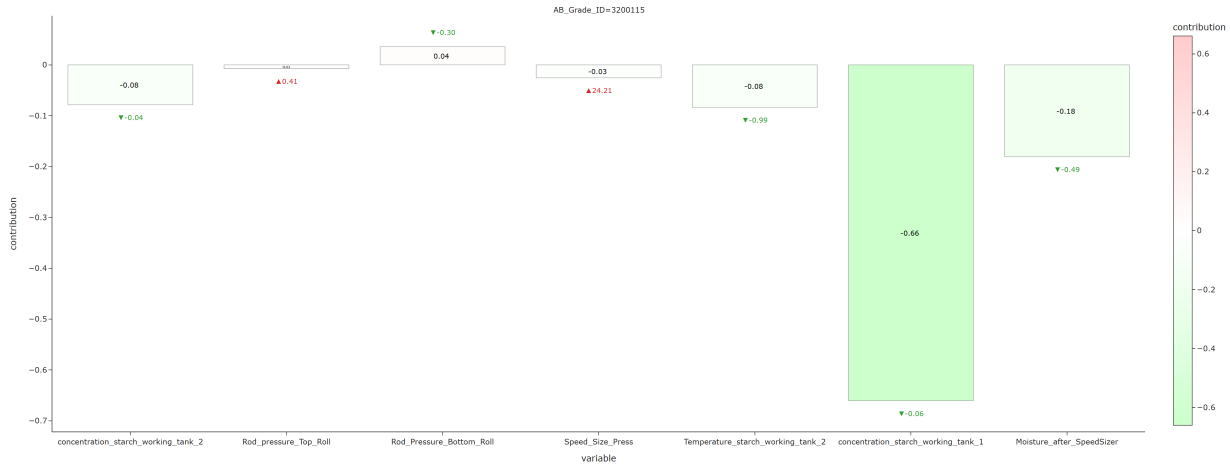


3.2.3 Electricity cost

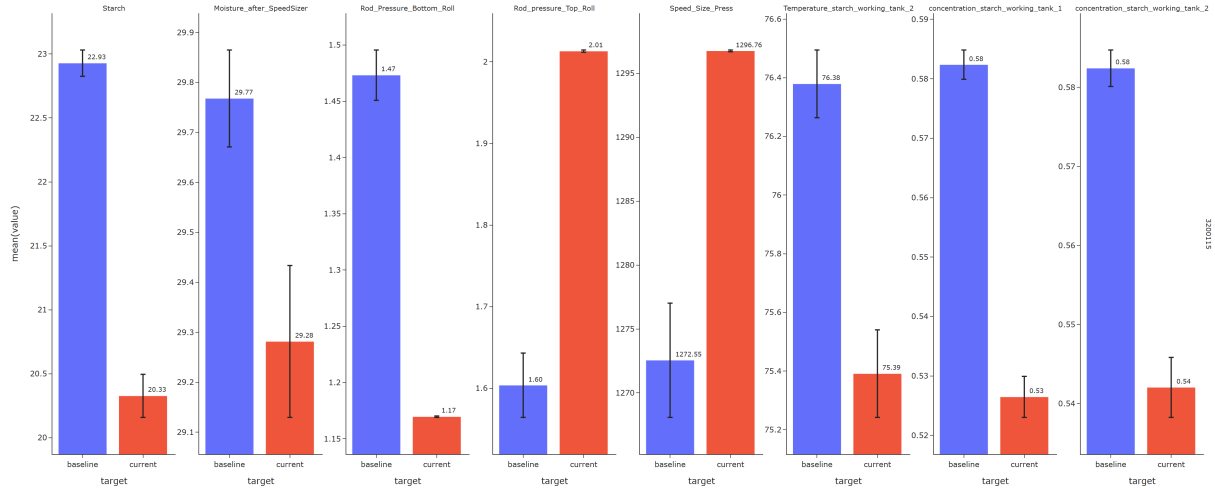
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Speed increased (+86.06) Cost-increasing drivers: - Lip settings increased (+0.28)



3.2.4 Starch cost



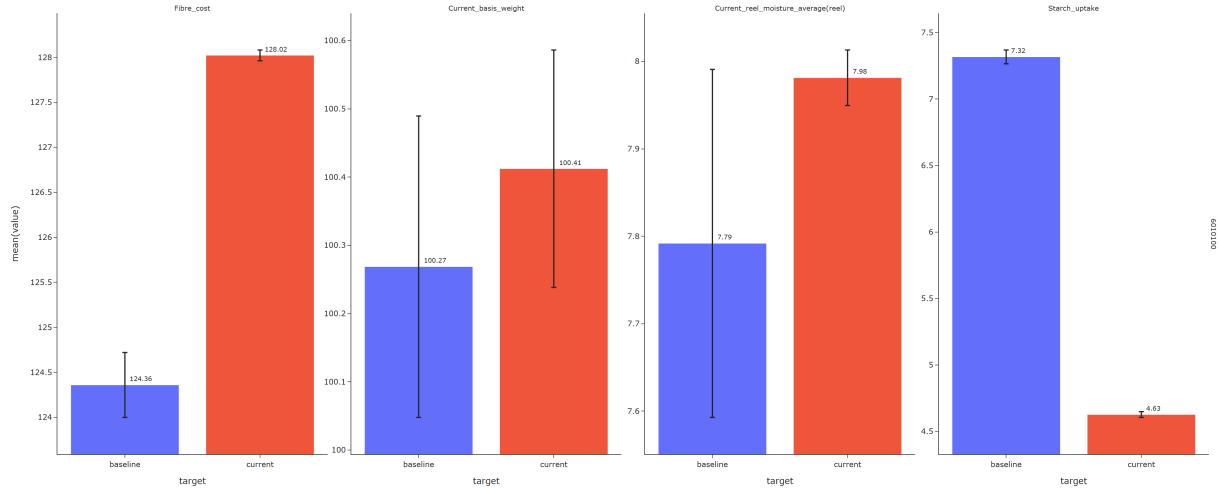
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Concentration starch working tank 1 decreased (-0.06) - Moisture after SpeedSizer decreased (-0.49)



3.3 6010100

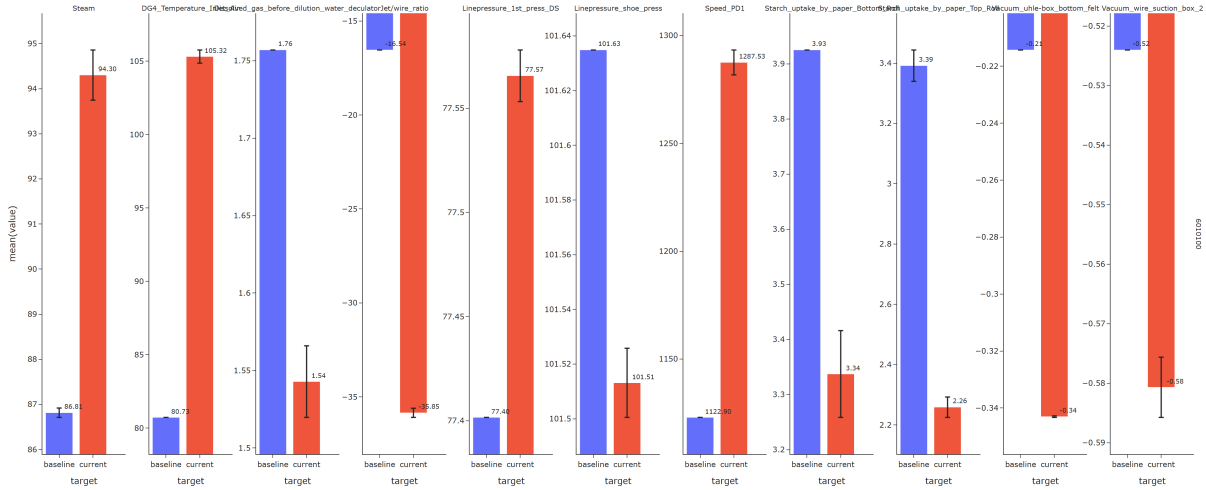
3.3.1 Fiber cost

For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-increasing drivers: - Starch uptake g/m2 decreased (-2.69)

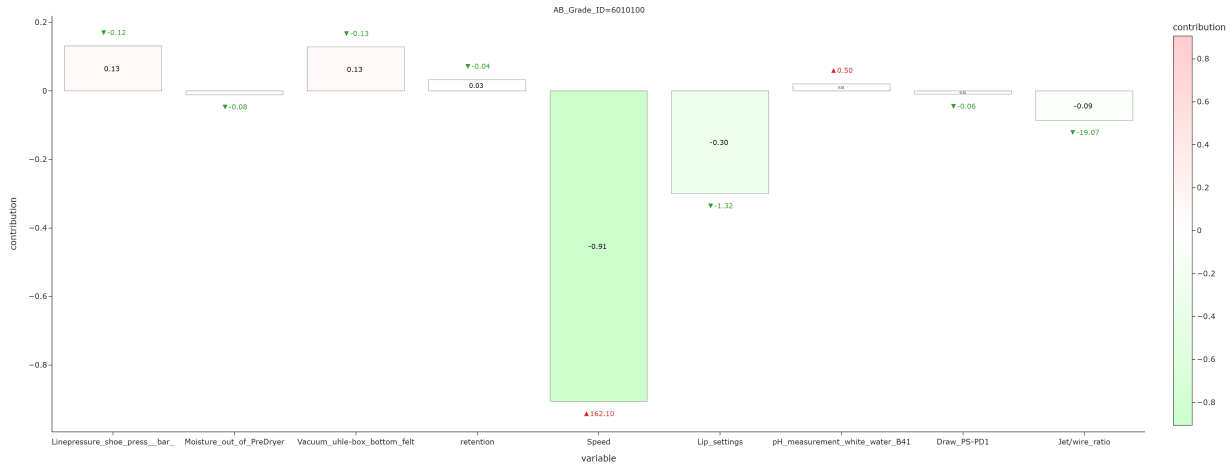


3.3.2 Steam cost

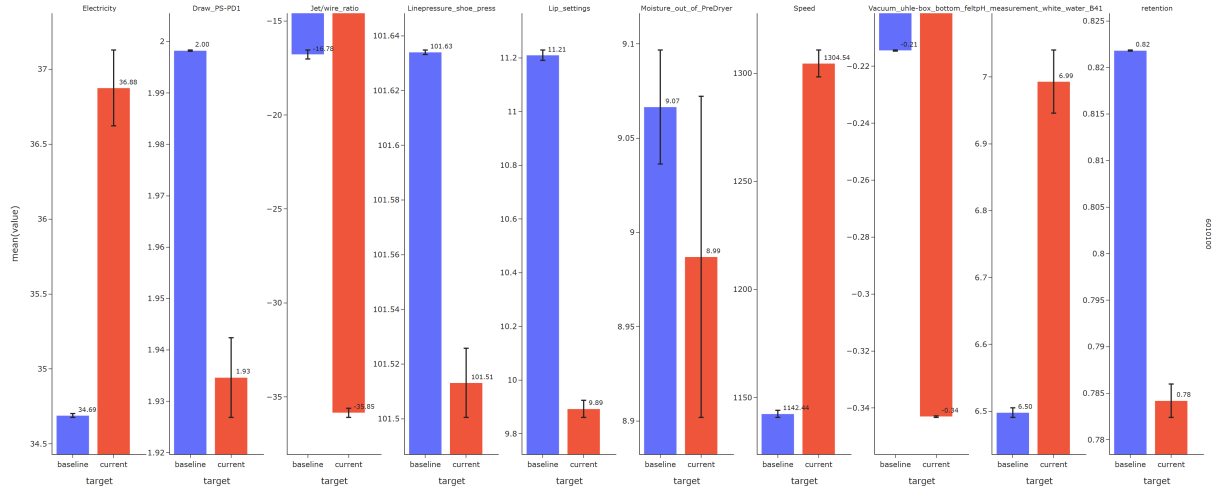
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Speed PD1 increased (+164.63) - Vacuum uhle-box bottom felt decreased (-0.13) - Starch uptake by paper Top Roll g/m2 decreased (-1.13) - Starch uptake by paper Bottom Roll g/m2 decreased (-0.59) Cost-increasing drivers: - Linepressure shoe press bar decreased (-0.12) - DG4 Temperature Inlet Air increased (+24.59)



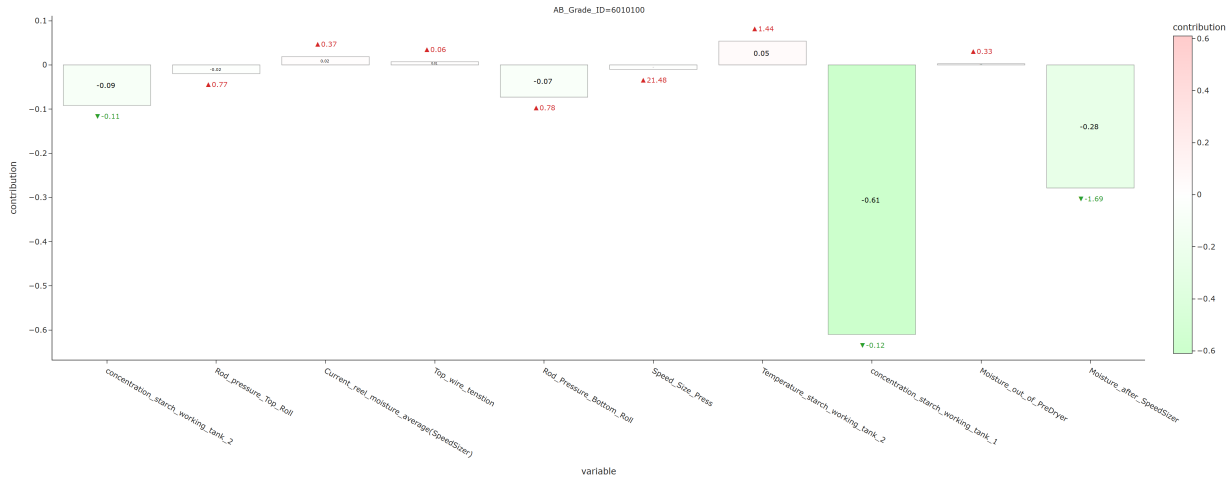
3.3.3 Electricity cost



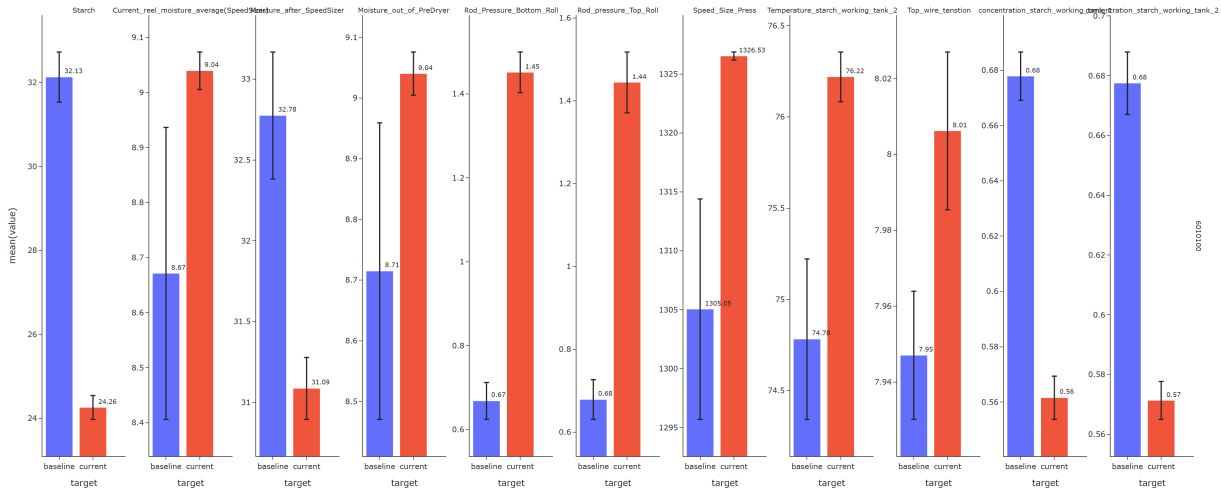
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Speed increased (+162.10) - Lip settings decreased (-1.32)



3.3.4 Starch cost

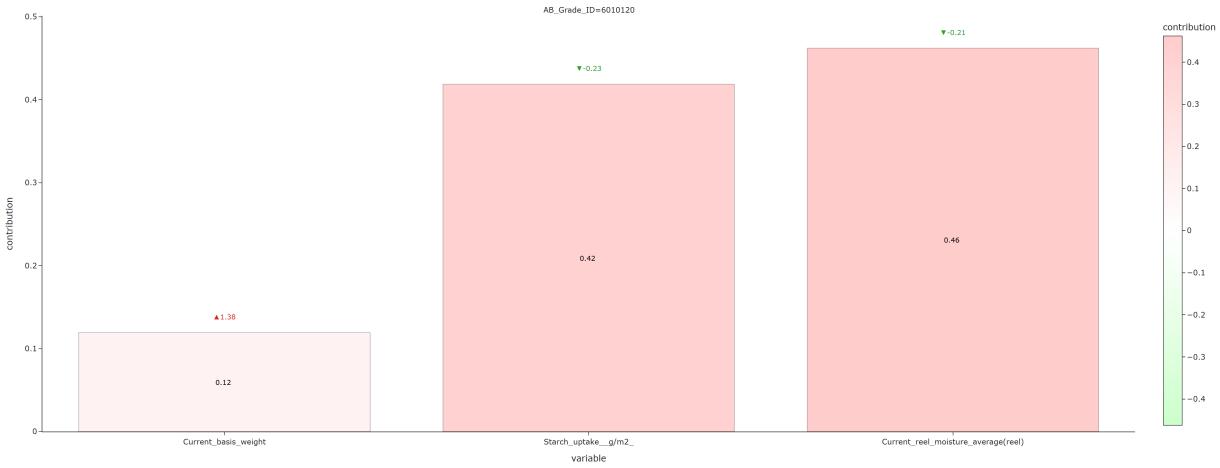


For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Concentration starch working tank 1 decreased (-0.12) - Moisture after SpeedSizer decreased (-1.69)

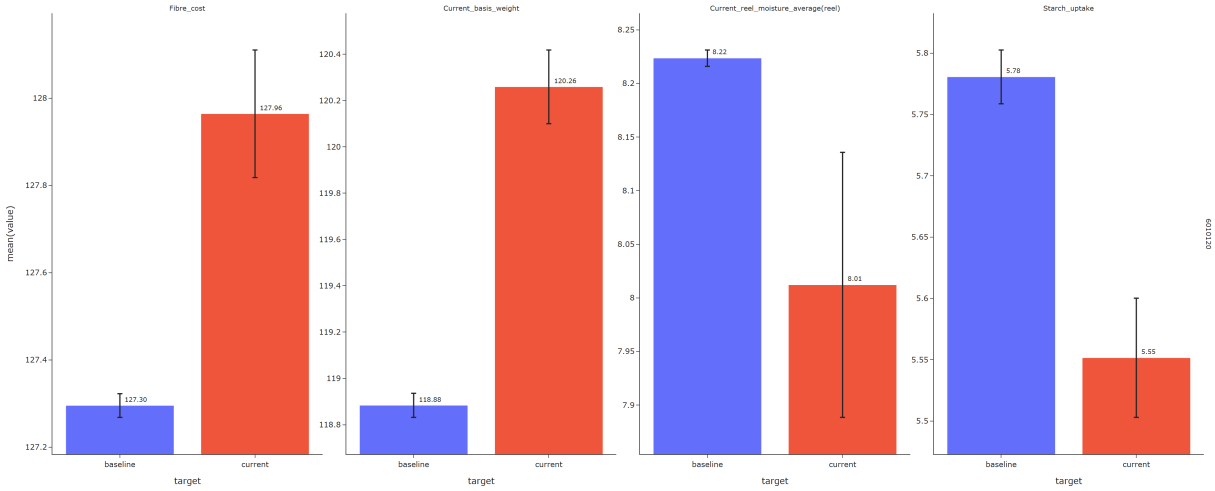


3.4 6010120

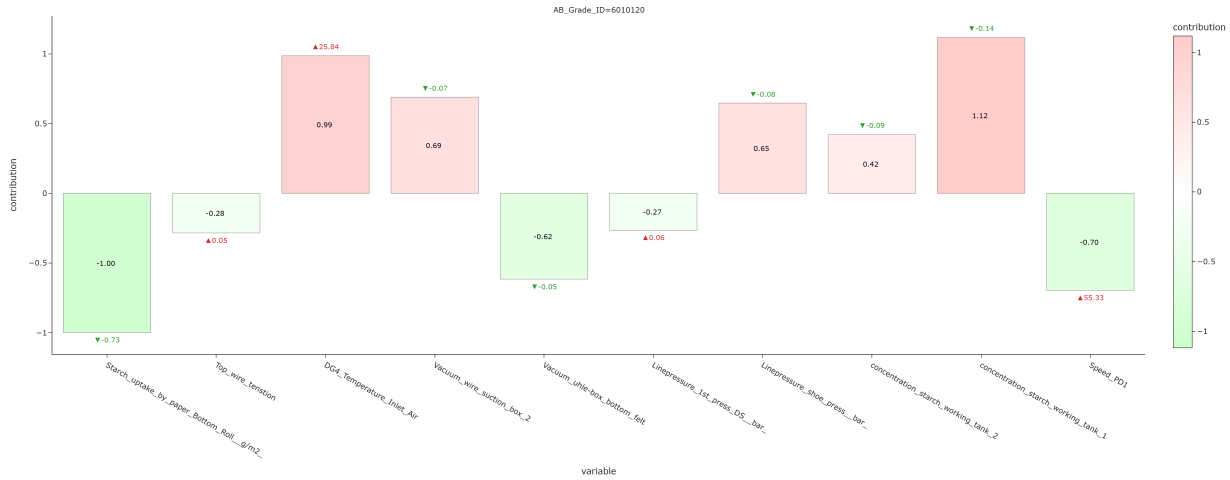
3.4.1 Fiber cost



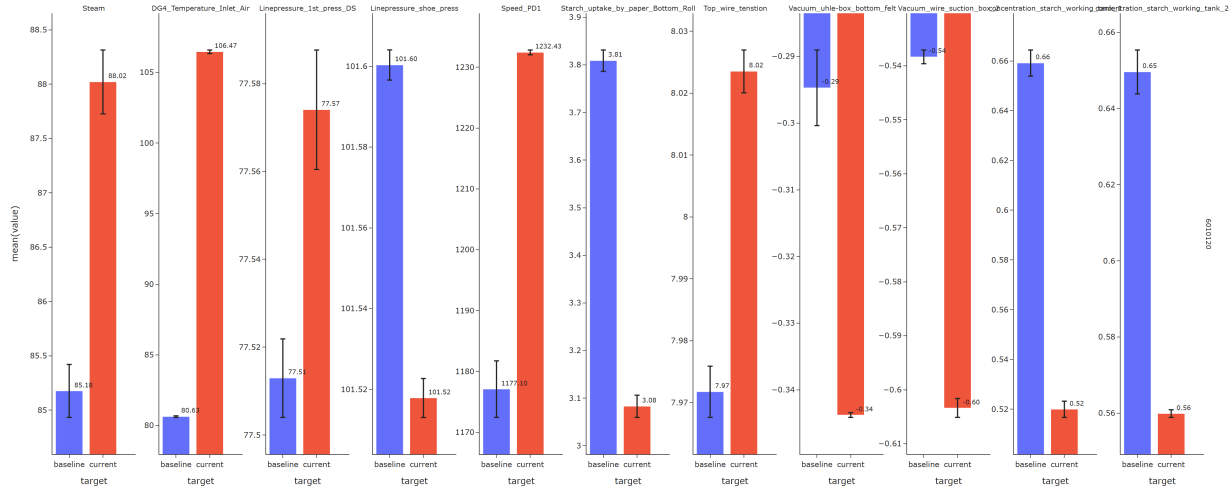
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-increasing drivers: - Current reel moisture average(reel) decreased (-0.21)



3.4.2 Steam cost

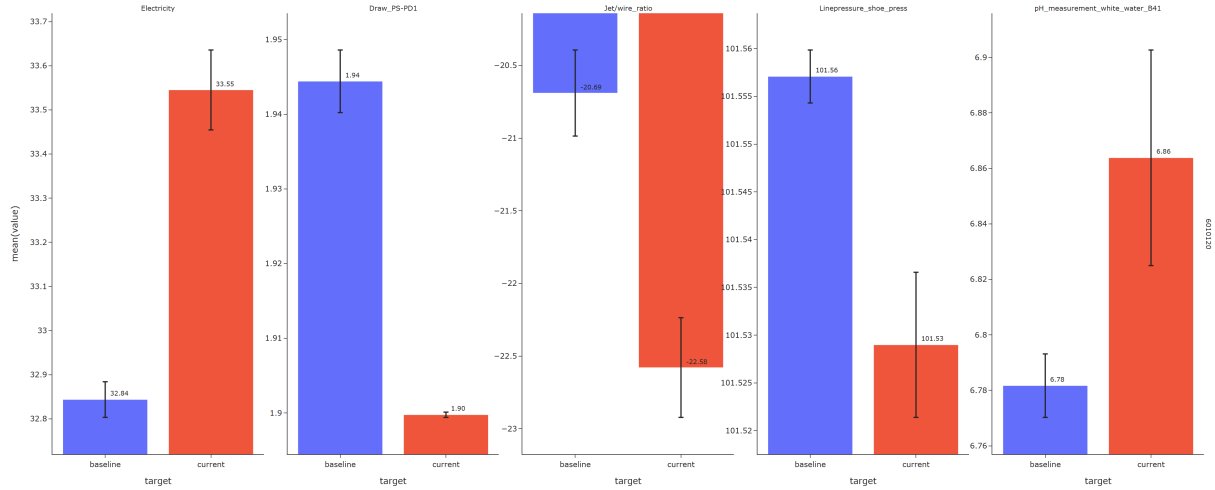


For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Starch uptake by paper Bottom Roll g/m2 decreased (-0.73) - Speed PD1 increased (+55.33) Cost-increasing drivers: - Concentration starch working tank 1 decreased (-0.14) - DG4 Temperature Inlet Air increased (+25.84) - Vacuum wire suction box 2 decreased (-0.07)



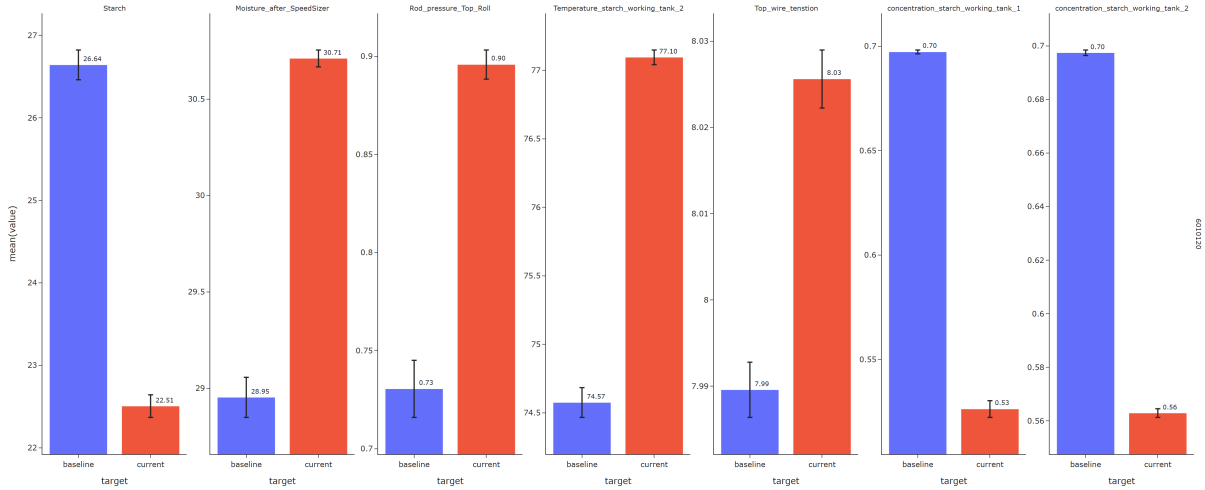
3.4.3 Electricity cost

For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-increasing drivers: - Linepressure shoe press bar decreased (-0.03)



3.4.4 Starch cost

For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Concentration starch working tank 1 decreased (-0.17) Cost-increasing drivers: - Moisture after SpeedSizer increased (+1.76)

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4 SUMMARY, DIAGNOSIS AND ACTION PLAN

4.1 Executive Summary

Overall performance shows positive cost reduction of **1.62 €/t** (resulting in 279.25 €/t), driven by process improvements (-2.76 €/t) that more than offset the impact of a more expensive grade mix (+1.14 €/t). However, overprocessing increased by **0.56%** (to 7.23%), indicating quality targets are being exceeded unnecessarily. Grade 6010085 shows the strongest cost improvement (-5.62

€/t), while grade 6010100 exhibits concerning overprocessing (+3.52%) and requires immediate attention.

4.2 Diagnosis by Functional Area

4.2.1 Cost Performance Analysis

Overall Trends: - **Starch consumption** (Size Press section): Largest cost reduction driver (-1.17 €/t average) [data_glossary.csv:e633305a49e7e154:1:0] - **Chemicals** (Cost section): Significant decrease (-1.07 €/t average) - **Steam** (Pre Dryier & After Dryier sections): Reduced consumption (-1.07 €/t average) - **Fibre cost** (Cost section): Only increasing component (+0.60 €/t average) - **Electricity** (Pre Dryier section): Minimal change (-0.01 €/t)

Grade-Specific Performance:

Grade 6010085 (Best performer: -5.62 €/t): - **Size Press**: Starch uptake decreased (-0.86 g/m²), reducing costs - **After Dryier**: DG4 inlet air temperature decreased (-12.03°C), improving steam efficiency - **Press Section**: Dewatering top wire suction box zone 2 increased significantly (+176.09), indicating improved water removal - **Wet End**: Dissolved gas before dilution water deculator increased (+0.25), potentially improving formation

Grade 3200115 (-2.29 €/t): - **Scanner section**: Current reel moisture decreased (-0.18%), indicating drier sheet requiring more energy - **Size Press**: Starch uptake bottom roll decreased (-0.33 g/m²), reducing chemical costs - **After Dryier**: DG1 inlet air temperature decreased (-10.39°C) - **Scanner**: Speed increased (+86.06 m/min), improving productivity [papermaking_science_technology.pdf:d6f6f20ccb65b658:7029:0]

Grade 6010100 (-1.60 €/t, but concerning overprocessing): - **Size Press**: Total starch uptake decreased significantly (-2.69 g/m²), but individual roll applications also decreased - **Scanner**: Speed increased substantially (+162.10-164.63 m/min) - **Press Section**: Vacuum uhle-box bottom felt decreased (-0.13), line pressure shoe press decreased (-0.12 bar) - **After Dryier**: DG4 temperature increased (+24.59°C), indicating higher energy input

Grade 6010120 (-1.41 €/t): - **Scanner**: Reel moisture decreased (-0.21%) - **Size Press**: Bottom roll starch uptake decreased (-0.73 g/m²) - **Scanner**: Speed increased (+55.33 m/min) - **After Dryier**: DG4 temperature increased (+25.84°C)

4.2.2 Quality Performance (Overprocessing Analysis)

Critical Issues:

Grade 6010100 (Highest overprocessing: 12.81%, +3.52%): - **CMT30** (compression strength): Increased +4.23%, indicating excessive strength - This suggests over-refining or excessive chemical addition in the Wet End/Size Press sections - Speed increases may have triggered compensatory over-addition of strength agents

Grade 6010120 (+0.90% to 5.56%): - **CMT30**: Increased +2.84% average - Moisture after SpeedSizer increased (+1.76%), indicating incomplete drying in Size Press section

Positive Developments:

Grade 3200115 (Reduced overprocessing: -1.95% to 6.81%): - **Burst strength**: Decreased -2.33%, bringing closer to target - Better optimization of strength properties

Grade 6010085 (Reduced overprocessing: -0.99% to 6.32%): - Improved balance between cost and quality

4.2.3 Process Section Analysis

Size Press Section [data_glossary.csv:e633305a49e7e154:1:0]: - Starch concentration in working tank 1 decreased across multiple grades (-0.06 to -0.17%) - Moisture after SpeedSizer generally decreased (except grade 6010120: +1.76%) - Starch uptake reductions driving major cost savings - Potential issue: Reduced starch may be compensated by over-specification in other areas

Drying Sections (Pre Dryier & After Dryier): - Inconsistent temperature management: DG4 temperatures decreased for grades 6010085/3200115 but increased for grades 6010100/6010120 - Steam pressure optimization opportunities exist [papermaking_science_technology.pdf:d6f6f20ccb65b658:4601:0] - Cylinder differential pressures show variability, indicating condensate removal issues

Press Section: - Vacuum settings show mixed changes across grades - Dewatering efficiency improvements in grade 6010085 (+176.09 suction box zone 2) - Shoe press line pressure reductions may be impacting downstream drying requirements

Scanner Section [data_glossary.csv:e633305a49e7e154:1:0]: - Speed increases across all grades (positive for productivity) - Moisture variations indicate grade change challenges [papermaking_science_technology.pdf:d6f6f20ccb65b658:7053:0] - Current reel moisture decreases may be driving higher energy costs

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4.3 Root Cause Analysis

4.3.1 Primary Issues:

1. **Grade Change Management**: The data suggests grade transitions are not optimized. Speed changes, moisture targets, and chemical dosing are not properly coordinated [papermaking_science_technology.pdf:d6f6f20ccb65b658:4601:0]. The context indicates "grade changes are demanding in that many actions must be taken simultaneously and coherently."

2. **Overprocessing in Grade 6010100**: CMT30 exceeding target by 4.23% while costs decreased suggests:

   - Compensation for reduced starch with other strength mechanisms
   - Inadequate quality control feedback loops
   - Possible over-refining in stock preparation (not visible in this data)

3. **Drying Section Inconsistency**: Temperature management varies significantly between grades, suggesting:

- Lack of standardized drying curves
- Reactive rather than predictive control
- Potential steam system bottlenecks [papermaking_science_technology.pdf:d6f6f20ccb65b658:4601:0]

4. **Fibre Cost Increase**: +0.60 €/t average increase (highest in grade 6010100: +0.70 €/t) may indicate:

   - Raw material price increases
   - Changes in furnish composition
   - Increased virgin fibre usage to compensate for reduced chemicals

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4.4 Action Plan

4.4.1 Immediate Actions (0-2 weeks)

Priority 1: Address Grade 6010100 Overprocessing - **Owner**: Process Engineer + Quality Manager - **Actions**: - Reduce CMT30 target or increase tolerance band - Decrease refining energy in stock preparation by 5-10% - Review and reduce retention aid dosing (Wet End chemicals) - Implement tighter moisture control after SpeedSizer (target reduction of 1.5%) - **Expected Impact**: Reduce overprocessing from 12.81% to <10%, save ~1.5 €/t - **Reference**: [papermaking_science_technology.pdf:d6f6f20ccb65b658:3395:0] - "1% increase in SCT causes operating cost increase of EUR 4.73/t"

Priority 2: Standardize Grade 6010120 Size Press Operation - **Owner**: Size Press Operator + Shift Supervisor - **Actions**: - Investigate moisture after SpeedSizer increase (+1.76%) - Check SpeedSizer nip pressure and roll temperatures - Verify starch concentration working tank 1 control loop - Reduce DG4 inlet air temperature by 10-15°C to match grade 6010085 performance - **Expected Impact**: Reduce moisture variability, save 0.3-0.5 €/t in steam costs

Priority 3: Optimize Drying Temperature Profiles - **Owner**: Drying Section Specialist - **Actions**: - Create standardized temperature setpoints for each grade based on grade 6010085 success - Implement feedforward control for DG4/DG1 temperatures based on incoming moisture - Review cylinder differential pressures for condensate removal efficiency - **Expected Impact**: Reduce steam consumption variability, potential 0.5 €/t savings

4.4.2 Short-term Actions (2-8 weeks)

Priority 4: Grade Change Optimization Program - **Owner**: Automation Engineer + Senior Process Engineer - **Actions**: - Implement dynamic grade change models [papermaking_science_technology.pdf:d6f6f20ccb65b658:4601:0] - Develop coordinated ramping schedules for speed, stock flow, and steam pressure - Create grade-specific transition matrices for chemical dosing - Install predictive moisture control during transitions - **Expected Impact**: Reduce grade change time by 15-20%, minimize transition broke - **Reference**: Context indicates "open-loop grade change systems are able to optimize the timing of these actions"

Priority 5: Starch Application Optimization - **Owner**: Size Press Engineer - **Actions**: - Analyze correlation between starch uptake reductions and quality parameters - Implement model-based starch dosing using moisture, speed, and basis weight inputs - Optimize starch concentration working tank 1 control (currently decreasing across grades) - Balance top roll vs. bottom roll application for each grade - **Expected Impact**: Maintain current starch savings (-1.17 €/t) while preventing overprocessing - **Data Source**: [data_glossary.csv:e633305a49e7e154:1:0] - Size Press variables

Priority 6: Press Section Dewatering Enhancement - **Owner**: Press Section Specialist - **Actions**: - Replicate grade 6010085 dewatering success (+176.09 suction box zone 2) to other grades - Optimize vacuum uhle-box bottom felt settings - Review shoe press line pressure strategy (currently decreasing) - Implement predictive maintenance for vacuum pumps (power monitoring) - **Expected Impact**: Reduce moisture to dryer section by 2-3%, save 0.8-1.2 €/t in steam

4.4.3 Medium-term Actions (2-6 months)

Priority 7: Integrated Cost-Quality Optimization System - **Owner**: Mill Technical Manager - **Actions**: - Implement real-time cost tracking by component (currently only periodic) - Develop quality-cost trade-off curves for each grade [papermaking_science_technology.pdf:d6f6f20ccb65b658:3395:0] - Create automated alerts for overprocessing >8% - Install statistical process control (SPC) for key cost drivers - **Expected Impact**: Prevent future overprocessing, optimize cost-quality balance - **Reference**: Context shows "1% increase in SCT causes EUR 4.73/t cost increase"

Priority 8: Fibre Cost Management - **Owner**: Fibre Line Manager + Purchasing - **Actions**: - Investigate root cause of +0.60 €/t fibre cost increase - Evaluate furnish composition alternatives (virgin vs. recycled ratios) - Assess impact of reduced chemicals on fibre requirements - Implement fibre cost tracking by grade - **Expected Impact**: Stabilize or reduce fibre costs by 0.3-0.5 €/t - **Data Source**: Cost section variables [data_glossary.csv:e633305a49e